01 / PROBLEM AND CONTRIBUTION

ASAP7 FinFET characterization

EEE 525 · VLSI Design · Lab 1 | Spring 2026

RESULT AT A GLANCE

Compared four ASAP7 threshold-voltage device options across -40, 25, and 120 °C, documenting transfer curves, subthreshold swing, and threshold-voltage extraction.

Engineering question

How do temperature and device threshold options affect FinFET transfer characteristics?

Situation

Selecting a transistor option requires understanding leakage, gate control, and drive behavior. The ASAP7 lab compares NMOS and PMOS devices across threshold options and temperature points.

Task

Run and interpret device-current sweeps, extract subthreshold swing and threshold voltage, and relate the reported curves to device-selection tradeoffs.

My contribution

Prepared the ASAP7 device characterization report with NMOS/PMOS curves, measurement-command screenshots, and temperature-dependent result tables.

Tools and concepts

FinFET characterization / ASAP7 PDK / SPICE simulation / Subthreshold swing / Temperature sweeps

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation

Compared SLVT, LVT, RVT, and SRAM model options, including LVT transfer curves at -40, 25, and 120 °C and multi-option comparisons at 25 °C.

02 / Implementation

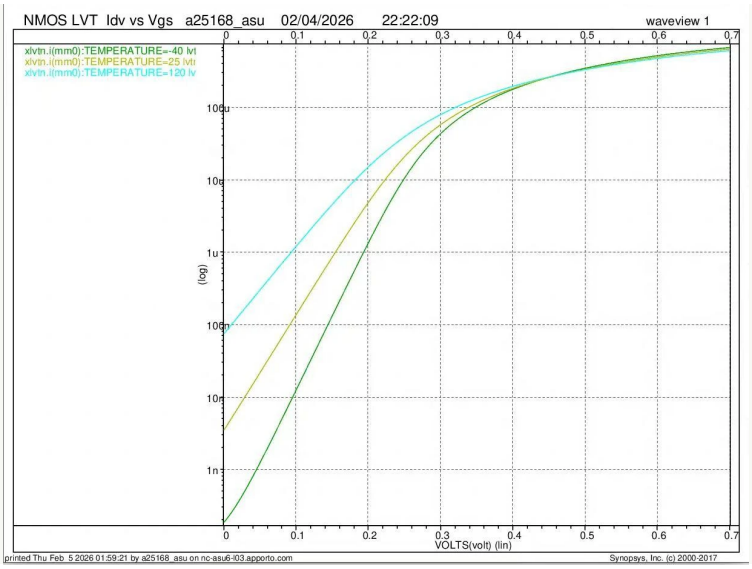
Tabulated subthreshold swing using the documented measurement expressions. NMOS LVT values are 49.166, 64.435, and 90.953 mV/dec at the three temperature points.

03 / Implementation

Documented threshold-voltage extraction and RVT output-current sweeps. Kept extraction results separate from a claim of measured cell speed or physical silicon behavior.

Subthreshold extraction

Subthreshold swing expresses the gate-voltage change needed for a decade change in current over an extraction interval. Reported values depend on the model, temperature, and chosen interval.



NMOS LVT temperature sweep

Source report p. 1: log current versus gate voltage at three temperatures.

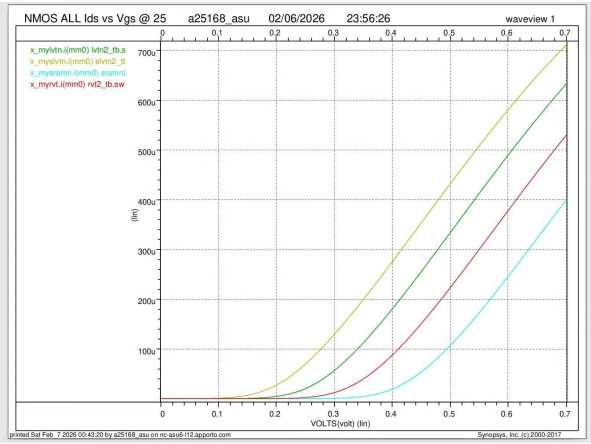
03 / RESULTS AND PERSPECTIVE

Evidence and engineering perspective

Threshold-voltage options	SLVT · LVT · RVT · SRAM
Temperature points	-40 / 25 / 120 °C
NMOS LVT swing at 25 °C	64.435 mV/dec

Reported result

The NMOS LVT subthreshold-swing table reports an increase from 49.166 mV/dec at -40 °C to 90.953 mV/dec at 120 °C, with 64.435 mV/dec at 25 °C. The report covers both polarities and four device options.



NMOS threshold-option comparison

Source report p. 2: SLVT, LVT, RVT, and SRAM transfer curves at 25 °C.

Scope and limitations

These are simulated device-model results. The report contains inconsistent fin arithmetic, PMOS ranking, and effective-current descriptions; those interpretations are excluded. PMOS signed slopes require a consistent magnitude convention before cross-polarity comparison.

Possible next steps - proposed work

- Correct the fin-width arithmetic and current-extraction descriptions in the source report.
- Use consistent slope magnitudes and extraction intervals for NMOS/PMOS comparisons.

Source material: EEE 525 VLSI Design Lab 1 submitted report, physical PDF pp. 1-9; subthreshold-swing results and measurement commands p. 5.